

Two-way variance of analysis problem statements of 5 types

Problem 1: Two-Way ANOVA WITH Interaction (Replicated)

Scenario: A researcher wants to study the effects of **Fertilizer Type** (Factor A: 3 levels) and **Planting Density** (Factor B: 2 levels) on **Crop Yield** (bushels per acre). The experiment is replicated twice (two plots per combination).

Problem Statement: Test at the 5% significance level ($p = 0.05$) whether the fertilizer type, planting density, or their interaction significantly affects crop yield.

Data Table (Yield in bushels/acre)

Fertilizer/Density	Density: Low	Density: High
Fertilizer 1	33.35, 31.45	28.25, 28.95
Fertilizer 2	32.60, 30.00	28.40, 27.70
Fertilizer 3	28.85, 28.10	28.50, 28.95

Problem 2: Two-Way ANOVA WITHOUT Interaction (Randomized Block Design)

Scenario: A manager wants to compare the effectiveness of three different **Teaching Methods** (Factor A) on student test scores. Because students are in three different **Schools** (Factor B, acting as a block), the manager uses a randomized block design to eliminate the variation caused by the school itself

Problem Statement: Test at the 5% significance level ($p = 0.05$) whether teaching methods differ in effectiveness, controlling for school differences.

Data Table (Average Test Scores)

School/Method	Method A	Method B	Method C
School 1	80	75	70
School 2	85	80	75
School 3	70	70	60

Problem 3: Two-Way ANOVA Without Replication

Scenario: A company wants to know if different sales programs (A, B, C) and different salesmen (1, 2, 3, 4) have a significant impact on weekly sales revenue.

- **Factor A (Row):** Salesmen (1, 2, 3, 4)
- **Factor B (Column):** Sales Programs (A, B, C)
- **Dependent Variable:** Sales (in \$1000s)

Note: There is only one observation per combination (e.g., Salesman 1 using Program A is only tested once), so interaction effects cannot be calculated.

Data Table:

Salesmen/Program	A	B	C
1	10	12	11
2	11	10	12
3	13	14	12
4	12	11	13

Problem Statement: Perform a two-way ANOVA at a 5% level of significance to test:

1. If there is a significant difference in sales among the four salesmen.
2. If there is a significant difference in sales among the three programs

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Problem 4: Two-Way ANOVA With Replication

Scenario: An agricultural researcher wants to know if fertilizer type (F1, F2) and watering frequency (Daily, Weekly) affect plant growth (in cm). To ensure accuracy, the experiment is repeated 3 times for each combination (3 replicates).

- **Factor A:** Fertilizer Type (F1, F2)
- **Factor B:** Watering Frequency (Daily, Weekly)
- **Dependent Variable:** Plant Growth (cm)
- **Replication:** n = 3 plants per treatment.

Data Table:

Fertilizer/Watering	Daily	Weekly
F1	15, 18, 17	10, 12, 11
F2	20, 22, 21	15, 16, 17

Problem Statement: Perform a two-way ANOVA at a 5% level of significance to test:

1. **Main Effect of Fertilizer:** Do fertilizers differ in growth?
2. **Main Effect of Watering:** Does frequency matter?
3. **Interaction Effect:** Does the effect of fertilizer depend on the watering frequency?

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Problem 5: Two-Way ANOVA Without Replication (Randomized Block Design)

Problem Statement: A company wants to test whether four different detergents (A, B, C, D) differ in their cleaning efficiency. They also suspect the engine model (1, 2, 3) affects the efficiency. To control for this, they test each detergent once on each engine model.

- **Factor 1 (Rows):** Detergent (A, B, C, D)
- **Factor 2 (Columns/Block):** Engine Model (1, 2, 3)

- **Dependent Variable:** Cleaning Score (0-100)
- **Goal:** Determine if detergent type significantly affects cleaning efficiency while controlling for engine model

Data Table (Cleaning Scores):

Detergent	Engine 1	Engine 2	Engine 3
A	45	47	40
B	30	33	35
C	50	52	48
D	35	38	32

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